5. Quantitative results

Table 1. Speech recognition accuracy by language pair

Language pair	Model	WER	CER	Mean latency*
Kiswahili-English	**Intron Sahara v2.5**	**0.4592**	0.2559	**3.08 s**
Kiswahili-English	**OpenAI Whisper Turbo**	0.5074	**0.1660**	25.60 s
Kiswahili-English	**Meta MMS**	0.7067	0.2759	12.51 s
Kiswahili-English	**Meta Omnilingual ASR**	0.7634	0.2790	200.80 s

*Latency is observed wall-clock time, not a controlled hardware benchmark: Sahara is hosted; Whisper/MMS were local; Omnilingual ran under WSL2/Linux on CPU.

Table 2. Downstream public-service form performance

Language pair	Model	Field exact	Exact fields	Complete forms
Kiswahili-English	**Intron Sahara v2.5**	0.2500	24/96	0/24
Kiswahili-English	**OpenAI Whisper Turbo**	**0.3021**	**29/96**	0/24
Kiswahili-English	**Meta MMS**	0.0625	6/96	0/24
Kiswahili-English	**Meta Omnilingual ASR**	0.0729	7/96	0/24

Key result: all four systems produced 0/24 completely correct forms. This is why SautiForm is designed as an assistive workflow with editable fields, clarification and explicit human confirmation rather than autonomous administrative submission.

6. Result interpretation

Model	What the numbers show
Sahara	Best WER (0.4592) and fastest observed latency (3.08 s). It recovered 24/96 exact fields. Lower WER did not translate into the best structured-field recovery.
Whisper Turbo	Best CER (0.1660) and strongest field exact match (29/96). WER was worse than Sahara and local CPU latency was substantially higher.
Meta MMS	WER/CER were materially weaker and only 6/96 fields were exact. Its Swahili adapter is monolingual, limiting within-utterance English handling.
Omnilingual ASR	7/96 exact fields, highest WER/CER and the slowest observed runtime in this CPU/WSL2 setup. It remains useful as a broad multilingual open comparator.

WER and CER are useful diagnostics, but the downstream tables are decisive for this application: a single wrong district, occupation, household count or certificate type can make an otherwise readable transcript unsuitable as a final administrative record.

7. Qualitative findings: strengths and weaknesses

Model / language	Strengths	Weaknesses / risks
Intron Sahara v2.5 Kiswahili-English	Native challenge backend for the target pair; best WER; fastest observed response; hosted API reduces local model-loading burden.	CER and field recovery trailed Whisper; 0/24 complete forms; hosted processing requires network access and appropriate user transparency.
Whisper Turbo Kiswahili-English	Best CER and strongest structured-field recovery; multilingual model handled mixed-language content better than the open monolingual comparator.	Slower local CPU inference; WER worse than Sahara; still 0/24 complete forms and therefore not safe for silent form acceptance.
Meta MMS Kiswahili-English	Open model; straightforward Swahili adapter; local inference was faster than Whisper in this evaluation.	The swh adapter is monolingual and does not explicitly model English switches; only 6/96 exact fields; 0/24 complete forms.
Meta Omnilingual ASR Kiswahili-English	Broad multilingual model family and an additional open comparison point.	Highest WER/CER here; only 7/96 exact fields; 200.80 s mean latency in the tested CPU/WSL2 runtime; more complex environment.

8. Product implication

Sahara is used in the live SautiForm product because it directly supports the required Kiswahili-English code-switching pair and produced the strongest WER/latency combination in this benchmark. However, the benchmark does **not** support autonomous form completion. SautiForm therefore shows the recognised transcript, extracts only supported fields, flags missing/invalid values, allows correction, reads the structured record back and requires explicit confirmation. The current prototype performs no government submission.

9. Limitations and reproducibility

Scope	The benchmark is small (24 clips; 0.06255 h), task-specific and limited to one language pair. It should not be interpreted as a general ranking of African ASR systems or as demographic fairness evidence.
Latency	Execution environments differ: Sahara is a hosted API; Whisper/MMS ran locally; Omnilingual ran under WSL2/Linux on CPU. Latency is operational evidence, not a controlled hardware comparison.
Privacy	Raw held-out audio and per-clip model outputs are not published in the public repository. Any optional public audio release should occur only where participant consent explicitly covers redistribution and the samples are de-identified.
Reproducibility	The repository contains manifest validation/freezing, model adapters, deterministic WER/CER and field metrics, the shared parser and benchmark runners. Aggregate values above are the frozen final results.

10. Submission-ready summary

Best WER / observed latency: Sahara (0.4592; 3.08 s) | Best CER / field recovery: Whisper Turbo (0.1660; 29/96 fields) | Complete forms: 0/24 for every model.

Repository: github.com/buriro-ezekia/sautiform-africa | Final held-out manifest hash: 794eddca2d656b176c0064dd7edd92da61b79266d113287de47247dc72a16448